

# SANJAY KUMAR

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## SUMMARY

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Mechatronics engineer with 16 months working on the development of autonomous field robots at Meraque Group, spanning STM32/FreeRTOS firmware, ~2 cm RTK-GPS positioning and a full digital twin of a production oil palm spraying AGV in NVIDIA Isaac Sim. Building things is what drew me to engineering in the first place, and it's still how I spend my own time, building robotics projects at home. Seeking a robotics or embedded systems role.

## EXPERIENCE

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- **Meraque Group Sdn Bhd** Malaysia  
*Robotics Engineer (Industry FYP Collaboration)* October 2025 – June 2026
  - Developed a digital twin of the company's oil palm spraying AGV in NVIDIA Isaac Sim, enabling virtual pre-deployment testing without plantation visits.
  - Delivered a validated simulation platform: 90.5% sim-to-real synchronisation accuracy, 96.3% tree detection rate, and 96.0% spray actuation accuracy across five structured test protocols.
- *Electronic Engineering Intern* February 2025 – September 2025
  - Built an RTK-GPS base station implementing FreeRTOS for multitask management, achieving ~2 cm positioning accuracy for autonomous AGV navigation.
  - Contributed to STM32 firmware development in C/C++ and supporting hardware integration for spraying AGVs.
  - Worked with the AI team to integrate perception outputs with low-level motor control over CAN/UART, closing the loop between detection and actuation for autonomous spraying.
  - Supported field testing, customer demonstrations, and QA during commercial deployment of the spraying AGVs.
- **Asia Pacific University Engineering Lab** Malaysia  
*Senior Technical Assistant (Part-time)* November 2023 – June 2026
  - Maintained 30+ items of lab equipment (oscilloscopes, multimeters, power supplies, embedded kits) with near-zero downtime; supervised 20+ students per session on safety and equipment usage.

## KEY PROJECTS

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- **Digital Twin of an Oil Palm Spraying AGV** *Isaac Sim, Gaussian Splatting, YOLOv8*
  - Reconstructed a real plantation environment via Gaussian Splatting, achieving 3.32% mean geometric error against ground-truth field measurements.
  - Built the simulation stack around Meraque's YOLOv8-seg tree detection model, adding a precision sprayer controller and real-time physical-virtual state synchronisation between the AGV and its digital twin.
- **Wall-E Companion Robot** *ROS2, CNN-BiGRU, DeepFace, Mistral LLM, Python*
  - Integrated a humanoid robot stack under a unified ROS2 architecture, fusing CNN-BiGRU speech-emotion recognition (89% accuracy), DeepFace vision, and a Mistral LLM for real-time child interaction.
  - Engineered a neck mechanism (DC motor, worm gear, rotary encoder) with a P-controller for face tracking.

## EDUCATION

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- **Asia Pacific University** (APU-DMU Dual Degree Scheme) Kuala Lumpur, Malaysia  
*BEng (Hons) Mechatronic Engineering | MEng, De Montfort University, UK* 2021 – June 2026

## LEADERSHIP

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- **IMechE APU Student Chapter** Malaysia  
*Vice-President* August 2024 – May 2026
  - Led a committee of 10+ members, directing planning and execution of 30+ technical workshops and competitions.

## TECHNICAL SKILLS

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- **Programming:** Python, C++, C.
- **Embedded & Automation:** ROS2, STM32, FreeRTOS, RTK-GPS, CAN/UART, PLC, Isaac Sim, YOLOv8.
- **Engineering Software & Tools:** MATLAB/Simulink, SolidWorks, LabVIEW, KiCad, LTSpice, Altair, Git, Linux.
- **Systems:** Hardware-Software Integration, Sensor Fusion, System Validation, QA.
- **Spoken:** English (Fluent), Urdu (Native), Hindi (Fluent).